Transmission Delay

- Time taken to put the data packet on the transmission link is called as **transmission delay**.
- Mathematically,
  - Transmission delay ∝ Length / Size of data packet
  - Transmission delay ∝ 1 / Bandwidth
- Thus,
  - **Transmission Delay**

Propagation Delay

- Time taken for one bit to travel from sender to receiver end of the link is called as **propagation delay**.
- Mathematically,
  - Propagation delay $\propto$ Distance between sender and receiver
  - Propagation delay $\propto$ 1 / transmission speed
- Thus,
  - **Propagation Delay**

Queuing Delay

- Time spent by the data packet waiting in the queue before it is taken for execution is called as **queuing delay.**
- It depends on the congestion in the network.

Processing Delay

- Time taken by the processor to process the data packet is called as **processing delay**.
- It depends on the speed of the processor.
- Processing of the data packet helps in detecting bit level errors that occurs during transmission.